MAT2002 **Name (Print):** _____
Spring 2026
HW2
Deadline: 4th March

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1. Solve the following equation:

   (i) $x^2y^3 + x\left(1+y^2\right)y' = 0$

   (ii) $\left(\frac{\sin y}{y} - 2e^{-x}\sin x\right) + \left(\frac{\cos y + 2e^{-x}\cos x}{y}\right)y' = 0$ (You can directly use: $\mu = ye^x$ as integrate factor)

Solution:

(i) (1) $y = 0$ is a solution.
(2) $y \neq 0$: $m(x,y) = x^2y^3, n(x,y) = x\left(1+y^2\right)$
$\frac{\partial m}{\partial y} = 3x^2y^2, \quad \frac{\partial n}{\partial x} = 1+y^2, \quad \frac{\partial m}{\partial y} \neq \frac{\partial n}{\partial x}$, not-exact!
multiply the $\mu(x,y)$ to the equation

$$x + \frac{1+y^2}{y^3}y' = 0$$
$$m^*(x,y) = x, n^*(x,y) = \frac{1+y^2}{y^3}, \quad \frac{\partial n^*}{\partial y} = \frac{\partial n^*}{\partial x} = 0, \text{ exact!}$$
$$xdx + \left(\frac{1}{y} + \frac{1}{y^3}\right)dy = 0$$
$$x^2 + \ln y^2 - \frac{1}{y^2} = C$$

(ii) multiply the $\mu(x,y)$ to the equation

$$\left(e^x \cdot \sin y - 2y\sin x\right) + ce^x\cos y + 2\cos x)\,y' = 0$$
$$m^*(x,y) = e^x\sin y - 2y\sin x, \quad n^*(x,y) = e^x\cos y + 2\cos x$$
$$\frac{\partial m^*}{\partial y} = e^x\cos y - 2\sin x, \quad \frac{\partial n^*}{\partial x} = e^x\cos y - 2\sin x$$
$$\frac{\partial m^*}{\partial y} = \frac{\partial n^*}{\partial x}, \quad \text{exact!}$$
$$\Psi(x,y) = \int m^*(x,y)dx + h(y) = e^x\sin y + 2y\cos x + h(y)$$
$$\frac{\partial \psi}{\partial y} = e^x\cos y + 2\cos x + h'(y) = n^* \Rightarrow h'(y) = 0$$
$$\Psi(x,y) = e^x\sin y + 2y\cos x = C$$

2. Show that if $\left(N_x - M_y\right)/(xM - yN) = R$, where R depends on the quantity xy only, then the differential equation

$$M + Ny' = 0$$

has an integrating factor of the form $\mu(xy)$. Find a general formula for this integrating factor.

Solution: Solution The ODE in question is

$$M(x,y) + N(x,y)y' = 0.$$

In general, this will not be an exact equation, so we will multiply both sides by the integrating factor $\mu = \mu(x,y)$ to make it so.

$$\mu M(x,y) + \mu N(x,y)y' = 0$$

Now that it is exact,

$$\frac{\partial}{\partial y}(\mu M) = \frac{\partial}{\partial x}(\mu N).$$

Expand both sides.

$$\frac{\partial \mu}{\partial y}M + \mu M_y = \frac{\partial \mu}{\partial x}N + \mu N_x$$

Assume now that μ is dependent on xy only: $\mu = \mu(xy)$.

$$x\mu'(xy)M + \mu M_y = y\mu'(xy)N + \mu N_x$$

Solve for $\mu'(xy)$.

$$\mu'(xy)(xM - yN) = (N_x - M_y)\,\mu$$
$$\mu'(xy) = \frac{N_x - M_y}{xM - yN}\mu$$

Divide both sides by μ.

$$\frac{\mu'(xy)}{\mu} = \frac{N_x - M_y}{xM - yN}$$

Let $z = xy$. The left side can be written as the derivative of a logarithm by the chain rule, and the right side is $R(z)$.

$$\frac{d}{dz}\ln\mu = R(z)$$

Integrate both sides with respect to z.

$$\ln\mu = \int R(z)dz$$

Therefore, exponentiating both sides, an integrating factor is

$$\mu(xy) = \exp\left[\int^{xy} R(z)dz\right].$$

3. Solve $\left(3x + \frac{6}{y}\right) + \left(\frac{x^2}{y} + 3\frac{y}{x}\right)\frac{dy}{dx} = 0$. (Hint: Use the result in Q2)

Solution:

This ODE is not exact at the moment because

$$\frac{\partial}{\partial y}\left(3x+\frac{6}{y}\right)=-\frac{6}{y^2}\neq\frac{\partial}{\partial x}\left(\frac{x^2}{y}+3\frac{y}{x}\right)=2\frac{x}{y}-3\frac{y}{x^2}$$

To solve it, we seek an integrating factor $\mu=\mu(x,y)$ such that when both sides are multiplied by it, the ODE becomes exact.

$$\mu\left(3x+\frac{6}{y}\right)+\mu\left(\frac{x^2}{y}+3\frac{y}{x}\right)\frac{dy}{dx}=0$$

Since the ODE is exact now,

$$\frac{\partial}{\partial y}\left[\mu\left(3x+\frac{6}{y}\right)\right]=\frac{\partial}{\partial x}\left[\mu\left(\frac{x^2}{y}+3\frac{y}{x}\right)\right]$$

Expand both sides.

$$\frac{\partial\mu}{\partial y}\left(3x+\frac{6}{y}\right)+\mu\left(-\frac{6}{y^2}\right)=\frac{\partial\mu}{\partial x}\left(\frac{x^2}{y}+3\frac{y}{x}\right)+\mu\left(2\frac{x}{y}-3\frac{y}{x^2}\right)$$

Following the hint, assume that μ is dependent on xy : $\mu=\mu(xy)$.

$$\begin{gathered}
x\mu'(xy)\left(3x+\frac{6}{y}\right)+\mu\left(-\frac{6}{y^2}\right)=y\mu'(xy)\left(\frac{x^2}{y}+3\frac{y}{x}\right)+\mu\left(2\frac{x}{y}-3\frac{y}{x^2}\right)\\
\mu'(xy)\left(3x^2+\frac{6x}{y}\right)=\mu'(xy)\left(x^2+\frac{3y^2}{x}\right)+\mu\left(2\frac{x}{y}-3\frac{y}{x^2}+\frac{6}{y^2}\right)\\
\mu'(xy)\left(2x^2+\frac{6x}{y}-\frac{3y^2}{x}\right)=\mu\left(\frac{2x}{y}-\frac{3y}{x^2}+\frac{6}{y^2}\right)\\
\mu'(xy)xy\left(\frac{2x}{y}+\frac{6}{y^2}-\frac{3y}{x^2}\right)=\mu\left(\frac{2x}{y}-\frac{3y}{x^2}+\frac{6}{y^2}\right)\\
\mu'(xy)xy=\mu
\end{gathered}$$

Let $z=xy$ and solve this ODE by separating variables.

$$\begin{gathered}
\frac{d\mu}{dz}z=\mu\\
\frac{d\mu}{\mu}=\frac{dz}{z}
\end{gathered}$$

Integrate both sides.

$$\ln\mu=\ln z+C$$

Exponentiate both sides.

$$\mu=(z)e^C$$

Taking e^C to be 1 , an integrating factor is

$$\mu = z = xy$$

Multiply both sides of the original ODE by xy.

$$\left(3x^2y + 6x\right) + \left(x^3 + 3y^2\right)\frac{dy}{dx} = 0$$

Because it's exact, there exists a potential function $\psi = \psi(x, y)$ that satisfies

$$\frac{\partial \psi}{\partial x} = 3x^2y + 6x$$
$$\frac{\partial \psi}{\partial y} = x^3 + 3y^2$$

Integrate both sides of equation partially with respect to x to get ψ.

$$\psi(x, y) = x^3y + 3x^2 + f(y)$$

Here $f(y)$ is an arbitrary function of y. Differentiate both sides with respect to y.

$$\psi_y(x, y) = x^3 + f'(y)$$

Comparing this to equation , we see that

$$f'(y) = 3y^2 \quad \rightarrow \quad f(y) = y^3.$$

As a result, a potential function is

$$\psi(x, y) = x^3y + 3x^2 + y^3.$$

Therefore,

$$\psi(x, y) = C_1$$
$$x^3y + 3x^2 + y^3 = C_1.$$

4. Determine a formula for the n-th Picard approximation for the initial value problem

$$\frac{\mathrm{d}y}{\mathrm{d}t} = ay, y(0) = 1, a \in \mathbb{R}$$

What is the limiting function $y(t) = \lim_{n\to\infty} y_n(t)$. Is it a solution? Are there other solutions that we may have missed?

Solution: The corresponding integral equation is $y(t) = 1 + \int_0^t ay(s)\mathrm{d}s$. Let $y_0 = y(0) = 1$,

we thus have

$$\begin{aligned}
y_0(t) &= y_0 = 1 \\
y_1(t) &= y_0 + \int_0^t a y_0(s)\mathrm{d}s = 1 + at \\
y_2(t) &= y_0 + \int_0^t a y_1(s)\mathrm{d}s = 1 + at + \frac{a^2t^2}{2} \\
y_3(t) &= y_0 + \int_0^t a y_2(s)\mathrm{d}s = 1 + at + \frac{a^2t^2}{2} + \frac{a^3t^3}{6} \\
&\vdots \\
y_n(t) &= y_0 + \int_0^t a y_{n-1}(s)\mathrm{d}s = 1 + at + \frac{a^2t^2}{2} + \cdots + \frac{a^nt^n}{n!}
\end{aligned}$$

We can write $y_n(t) = \sum_{k=0}^{n} \frac{a^k t^k}{k!}$. We recognize this sum as the first n terms of the Taylor series expansion for e^{at}. Thus the limiting function is $y(t) = \lim_{n\to\infty} y_n(t) = e^{at}$. It is straightforward to verify that it is a solution. Since $y' = ay$ is a linear first-order ODE, $p(t) = a$ is continuous in $(-\infty, +\infty)$. Thus, $y(t) = e^{at}$ is the only solution to the given initial value problem.

5. (i) Find the exact solution of the initial value problem

$$\frac{\mathrm{d}y}{\mathrm{d}t} = y^2, y(0) = 1.$$

(ii) Calculate the first three Picard approximations $y_1(t), y_2(t)$, and $y_3(t)$ and compare these results with the exact solution.

Solution:

The equation is separable, so separate the variables to get $y^{-2}\,\mathrm{d}y = \mathrm{d}t$. Integrating gives $-y^{-1} = t + c$, and the initial condition $y(0) = 1$ implies that the integration constant $c = -1$, so that the exact solution is

$$y(t) = \frac{1}{1-t}, t < 1$$

To apply Picard's method, let $y_0 = y(0) = 1$ and define

$$\begin{aligned}
y_0(t) &= y_0 = 1 \\
y_1(t) &= y_0 + \int_0^t (y_0(s))^2 \;\mathrm{d}s = 1 + t \\
y_2(t) &= y_0 + \int_0^t (y_1(s))^2 \;\mathrm{d}s = 1 + t + t^2 + \frac{t^3}{3} \\
y_3(t) &= y_0 + \int_0^t (y_2(s))^2 \;\mathrm{d}s = 1 + t + t^2 + t^3 + \frac{2t^4}{3} + \frac{t^5}{3} + \frac{t^6}{9} + \frac{t^7}{63}
\end{aligned}$$

For the exact solution, we have

$$y(t) = \frac{1}{1-t} = 1 + t + t^2 + t^3 + t^4 + \cdots$$

where $|t| < 1$. Comparing $y_3(t)$ to the exact solution, we see that the series agree up to order 3 when $|t| < 1$.

6. Find the Wronskian of the given pair of functions

   (i) $e^t \sin t, \quad e^t \cos t.$

   (ii) $\cos^2 \theta, \quad 1 + \cos 2\theta$

Solution:

$$W\left(e^t \sin t, e^t \cos t\right) = \begin{vmatrix} e^t \sin t & e^t \cos t \\ e^t(\sin t + \cos t) & e^t(\cos t - \sin t) \end{vmatrix} = -e^{2t}$$

$$W\left(\cos^2 \theta, 1 + \cos 2\theta\right) = \begin{vmatrix} \cos^2 \theta & 1 + \cos 2\theta \\ -2 \sin \theta \cos \theta & -2 \sin 2\theta \end{vmatrix} = 0.$$

7. For each of the following, determine the longest interval in which the given initial value problem is certain to have a unique twice-differentiable solution. Do not attempt to find the solution.

   (i) $y'' + (\cos t)y' + 3(\ln |t|)y = 0, \quad y(2) = 3, \quad y'(2) = 1.$

   (ii) $(x-3)y'' + xy' + (\ln |x|)y = 0, \quad y(1) = 0, \quad y'(1) = 1$

   (iii) $(x-2)y'' + y' + (x-2)(\tan x)y = 0, \quad y(3) = 1, \quad y'(3) = 2$

Solution:

(i) There is a point of discontinuity at $t = 0$, which means the interval in which the general solution is unique and twice-differentiable is either $-\infty < t < 0$ or $0 < t < \infty$. Because y and y' are prescribed at $t = 2$, the general solution is unique and twice-differentiable on $0 < t < \infty$.

(ii) Divide both sides of the ODE by $x - 3$ so that the coefficient of y'' is 1 .

$$y'' + \frac{x}{x-3}y' + \frac{\ln |x|}{x-3}y = 0$$

There are points of discontinuity at $x = 0$ and $x = 3$, which means the interval in which the general solution is unique and twice-differentiable is either $-\infty < x < 0$ or $0 < x < 3$ or $3 < x < \infty$. Because y and y' are prescribed at $x = 1$, the general solution is unique and twice-differentiable on $0 < x < 3$.

(iii) Divide both sides of the ODE by $x-2$ so that the coefficient of y'' is 1.

$$y'' + \frac{1}{x-2}y' + \frac{\sin x}{\cos x}y = 0$$

There are points of discontinuity at $x = 2$ and $x = (2n-1)\pi/2$ for $n = 0, \pm 1, \pm 2$. The general solution is unique and twice-differentiable in some interval between two adjacent points. Because y and y' are prescribed at $x = 3$, the general solution is unique and twice-differentiable on $2 < x < 3\pi/2$.

8. For each the following, find the Wronskian of two solutions of the given differential equation without solving the equation.

(i) $(\cos t)y'' + (\sin t)y' - ty = 0$

(ii) $x^2y'' + xy' + \left(x^2 - v^2\right)y = 0$, Bessel's equation

(iii) $\left(1-x^2\right)y'' - 2xy' + \alpha(\alpha+1)y = 0$, Legendre's equation

Solution:

(i) Writing the equation in standard form, we find that $P(t) = \sin t/\cos t$. Hence the Wronskian is $W(t) = ce^{-\int(\sin t/\cos t)dt} = ce^{\ln|\cos t|} = c\cos t$, in which c is some constant.

(ii) After writing the equation in standard form, we have $P(x) = 1/x$. The Wronskian is $W(x) = ce^{-\int(1/x)dx} = ce^{-\ln|x|} = c/x$, in which c is some constant.

(iii) Writing the equation in standard form, we find that $P(x) = -2x/\left(1-x^2\right)$. The Wronskian is $W(x) = ce^{-\int -2x/\left(1-x^2\right)dx} = ce^{-\ln\left|1-x^2\right|} = c/\left(1-x^2\right)$, in which c is some constant.